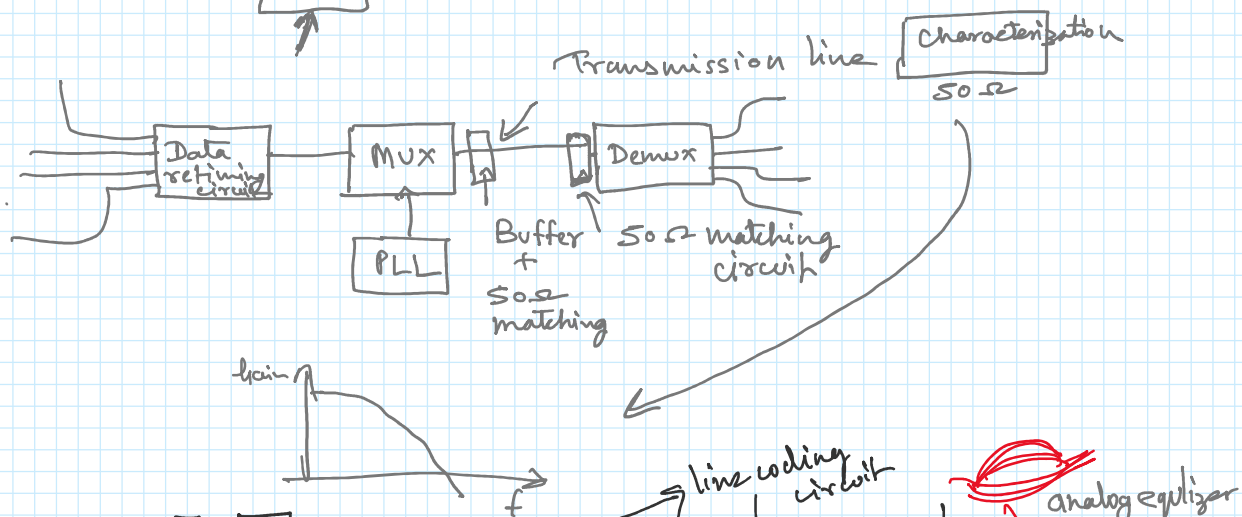
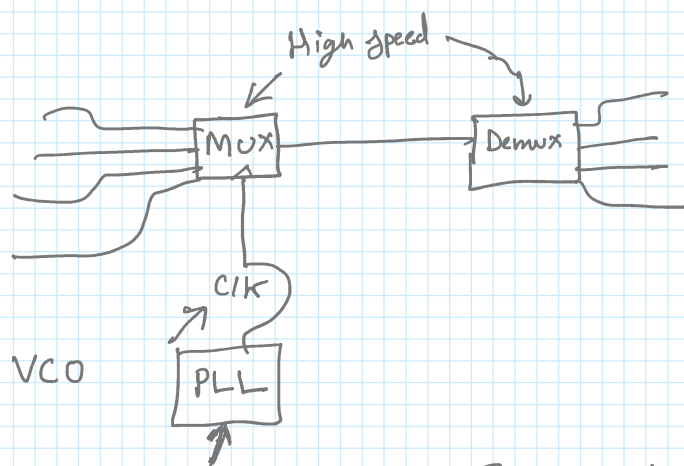
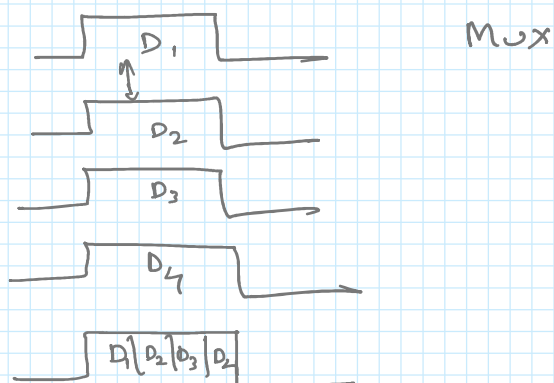
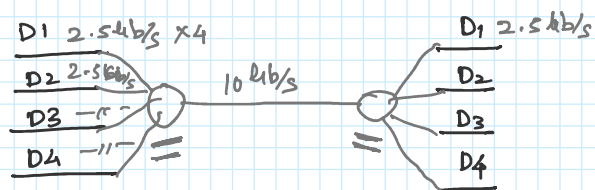
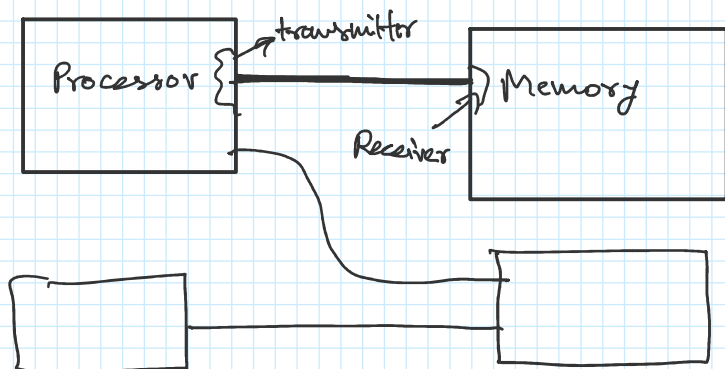
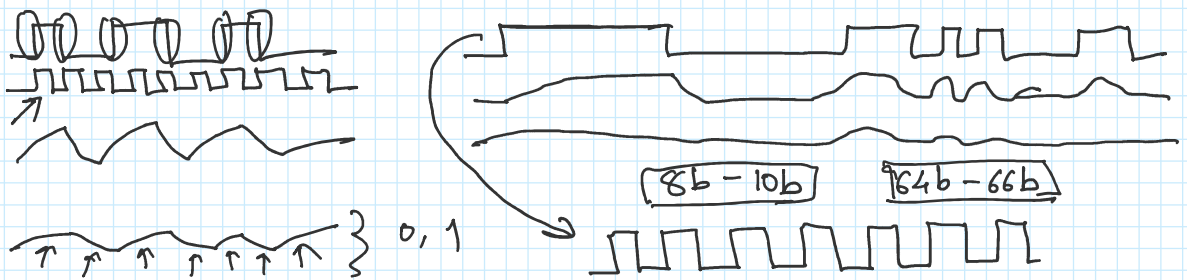
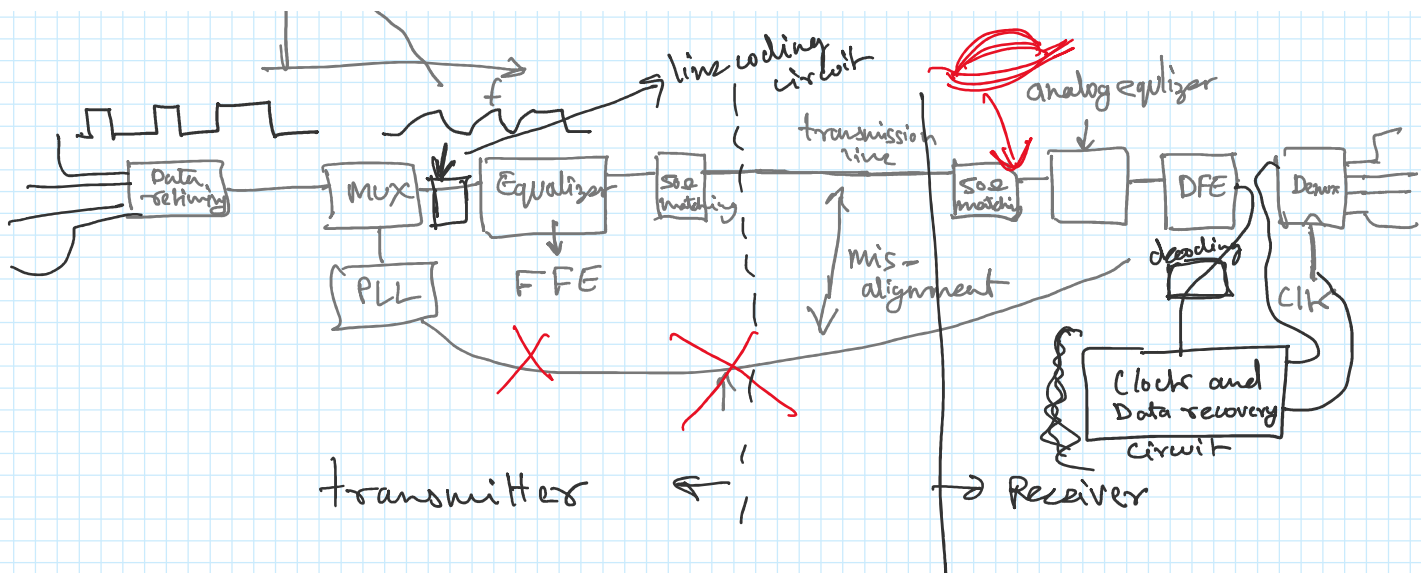
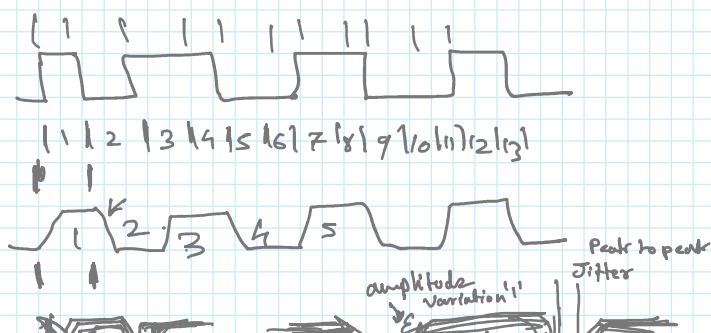
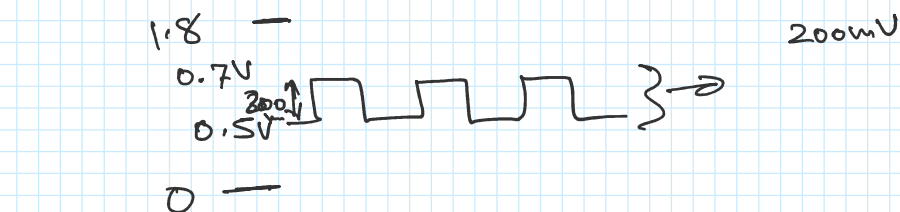
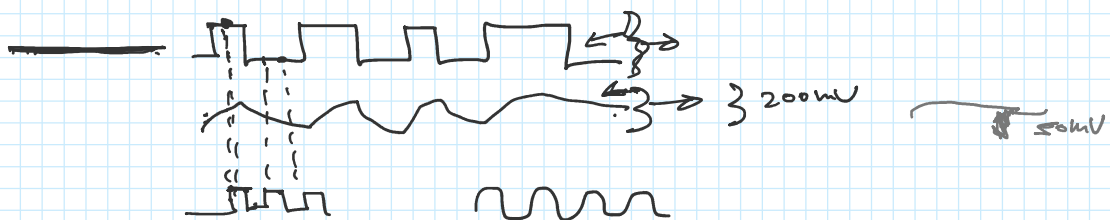


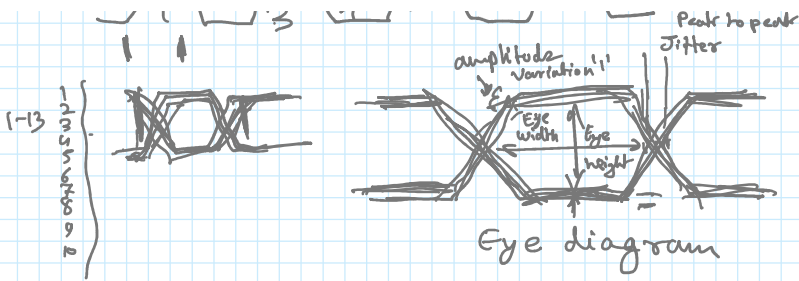
Clock and data recovery circuit





Clock and data recovery circuit.

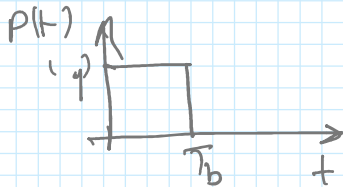




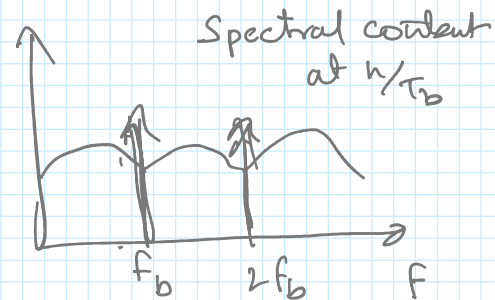
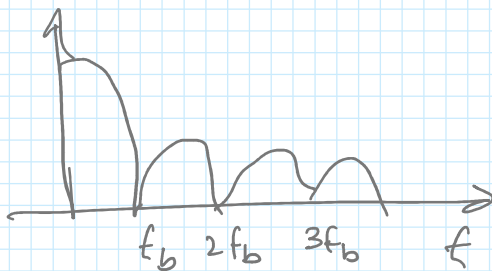
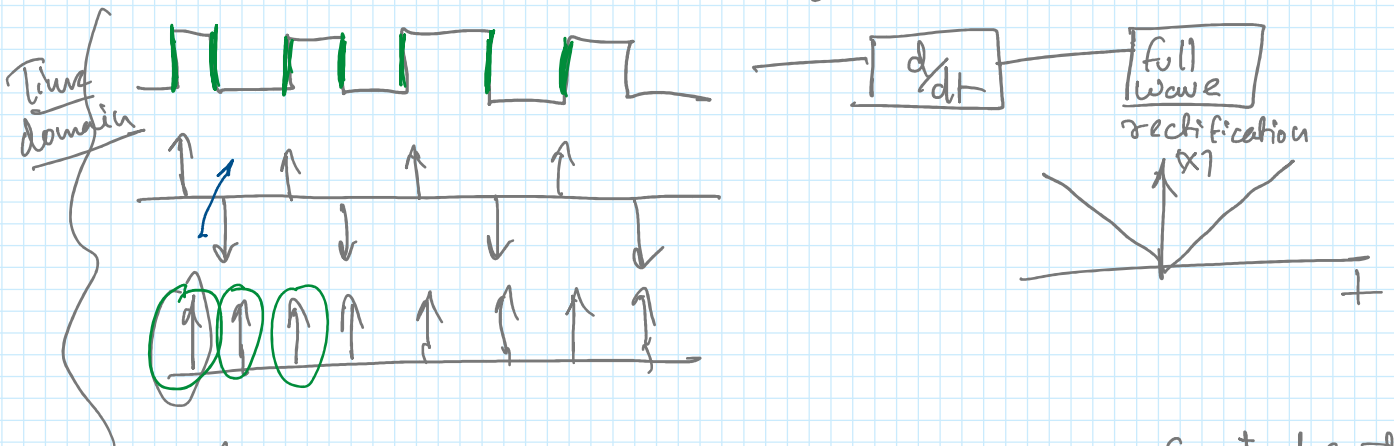
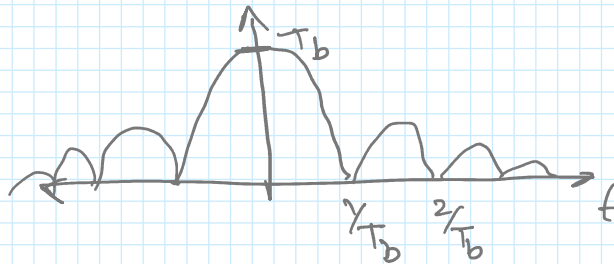
Detection of Data and clock



$$x(t) = \sum_k \underbrace{b_k}_{0 \text{ or } 1} \underbrace{P(t - kT_b)}_{\text{Shifted}}$$

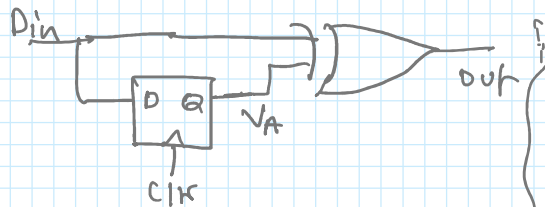
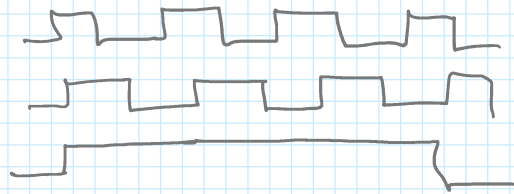
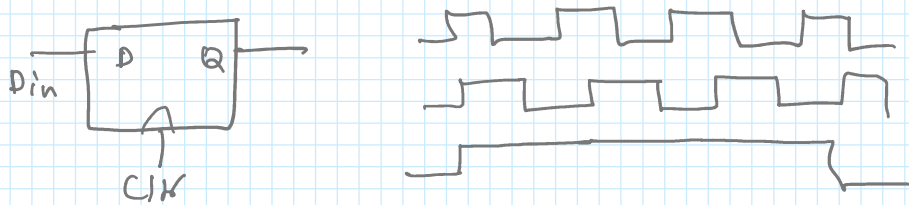
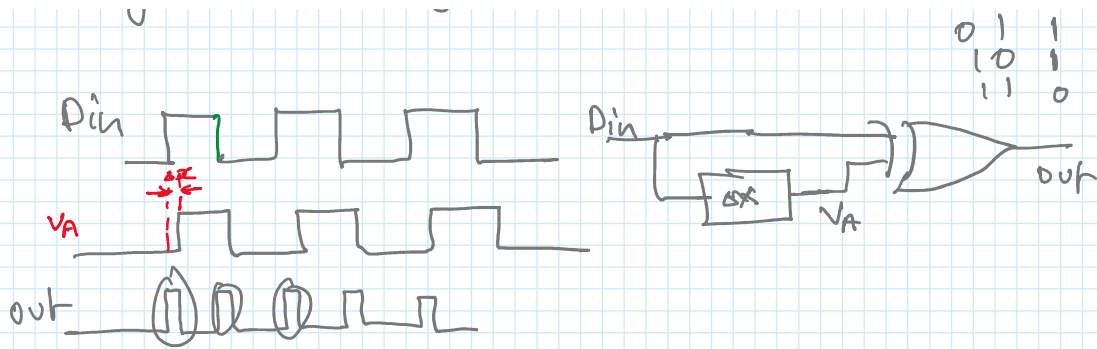


PSD : $S_x(f) = T_b \text{sinc}^2(f T_b)$

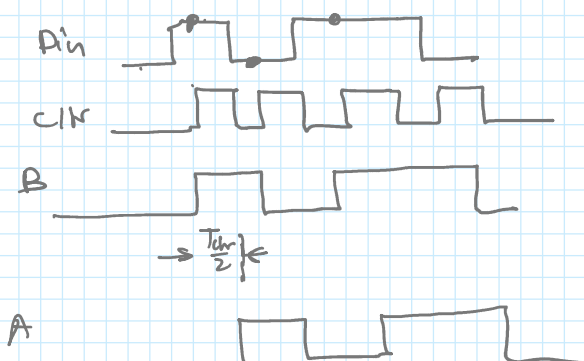
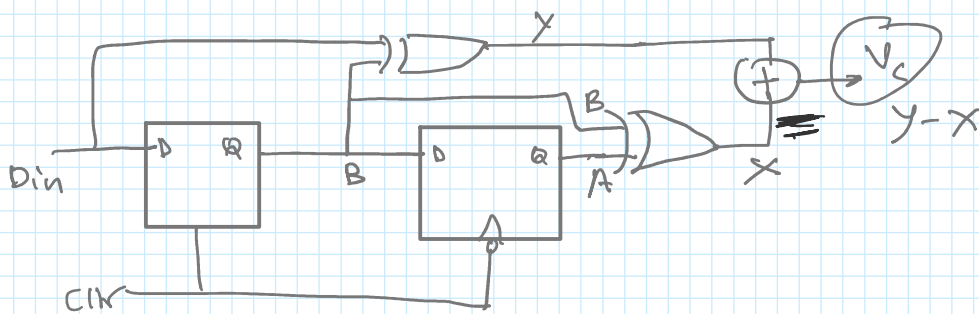
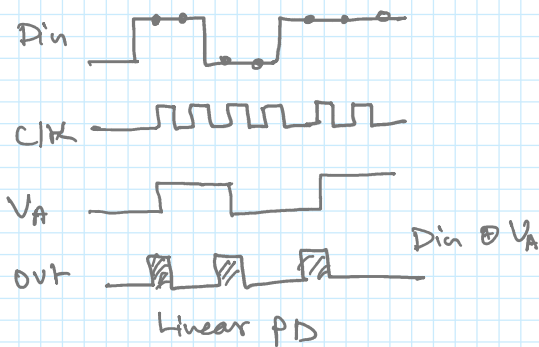


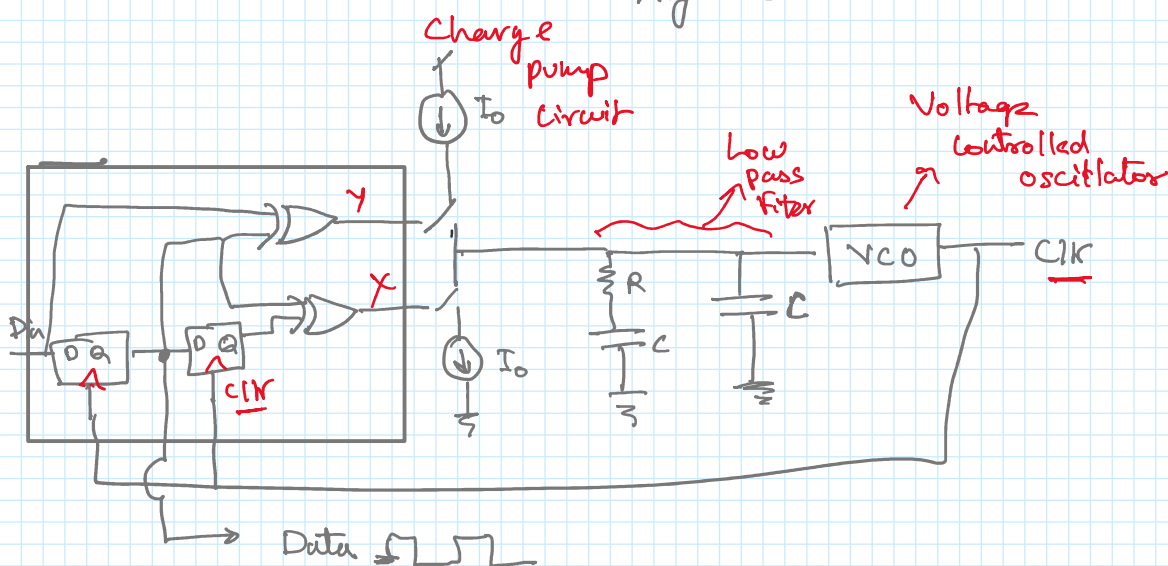
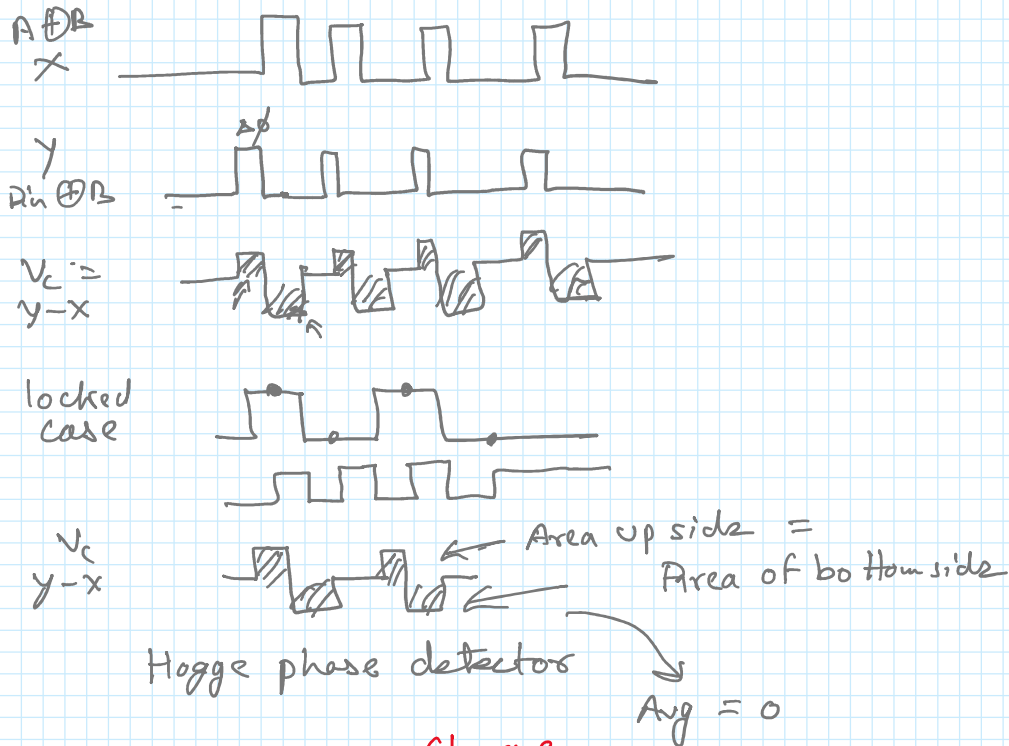
Edge detector :

0 0 0
0 1 1
1 0 1
1 1 0



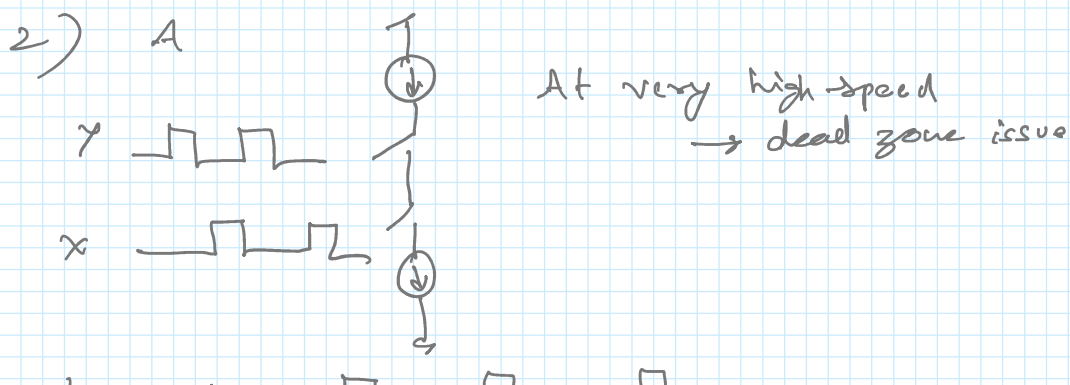
Data density
Arg value of out is function
of data density,



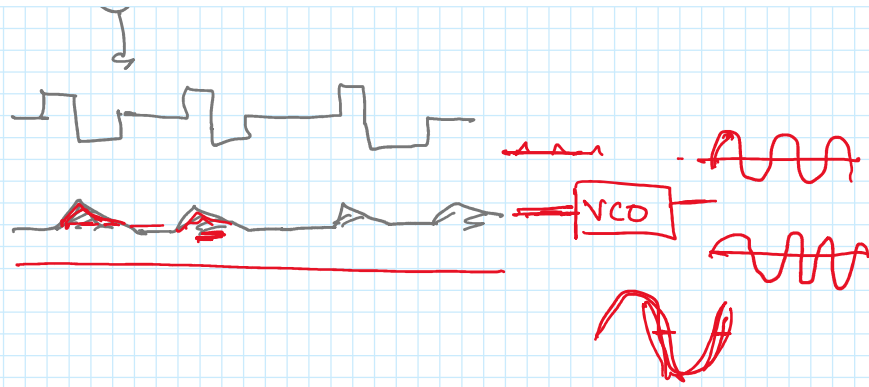


Type II single loop CDR circuit

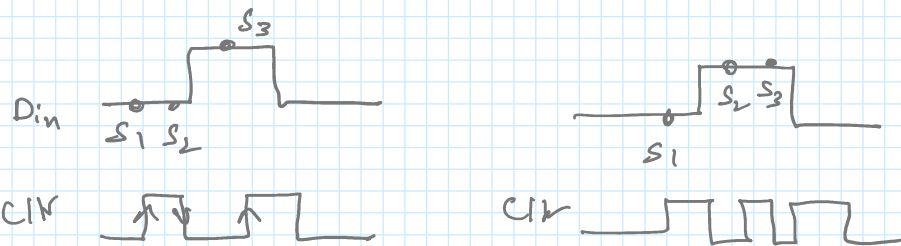
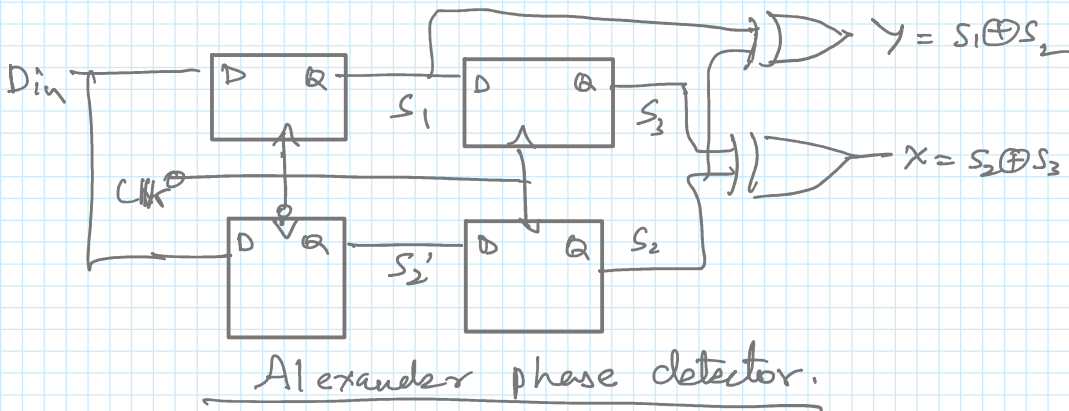
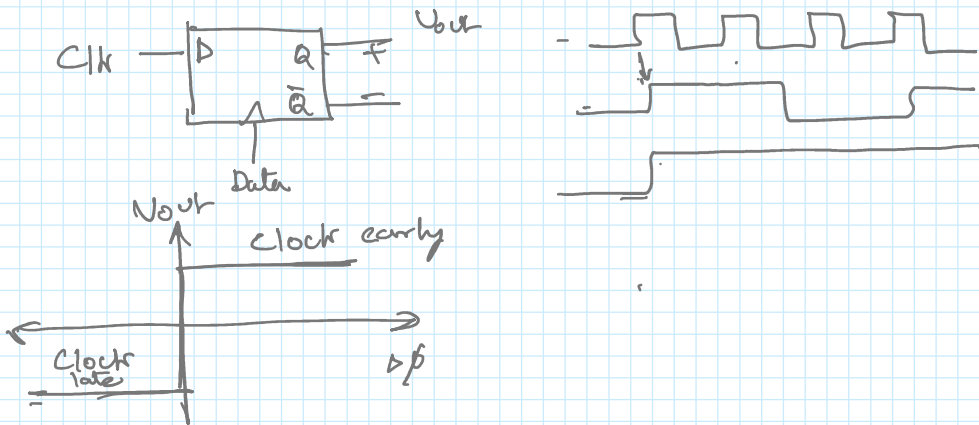
- 1) Finite delays in FFs $\rightarrow t_{cq}$
 - $\rightarrow$ changes $Y \rightarrow \Delta T \rightarrow t_{cq} \rightarrow$ phase error
- High speed ΔT could be a large fraction of T_{ck} degrades jitter tolerance.



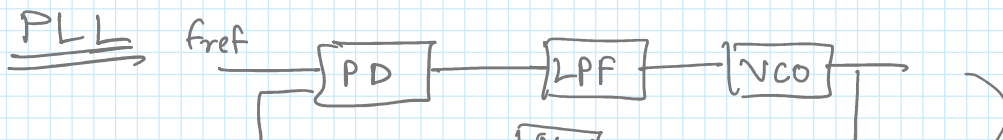
3/ y-x

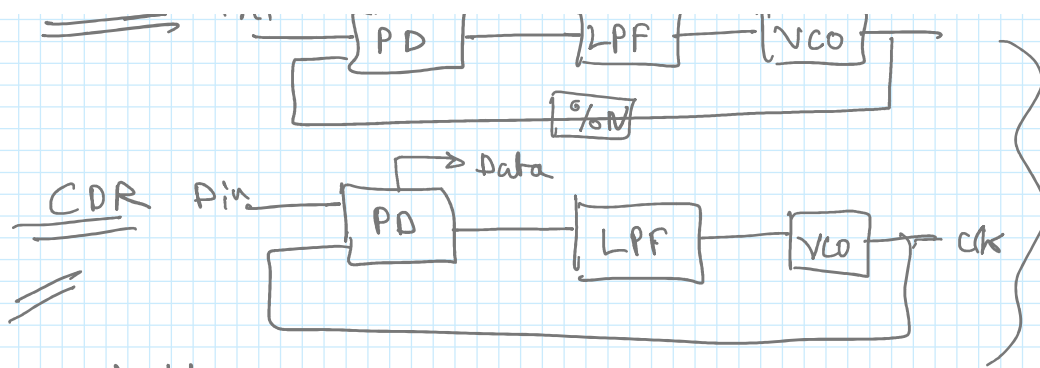


Binary PD



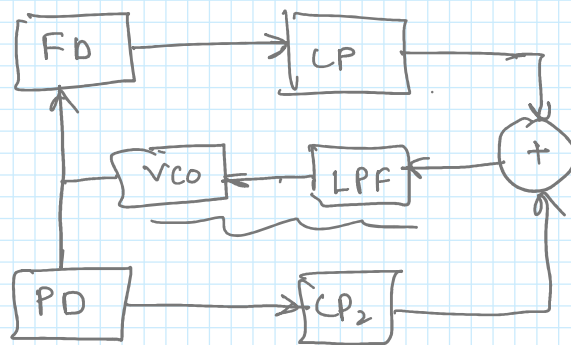
- $S_1 \oplus S_2 = 1$ & $S_1 \oplus S_3 = 0 \Rightarrow$ Clock is late.
- $S_1 \oplus S_2 = 0$ & $S_1 \oplus S_3 = 1 \Rightarrow$ Clock is early
- if $S_1 \oplus S_2 = S_2 \oplus S_3 \Rightarrow$ no data transition





→ Hogge

→ A PD



FLL + PLL with shared loop filter